

Students were asked the following questions on a survey:

- A. Do you have no siblings, one sibling, or two or more siblings?
- B. Do you have dogs, cats, or no pets?

The two-way table gives partial results. Use this information to complete questions 1-5.

	No Siblings	One Sibling	Two or More Siblings	Total
Dogs	8	15	7	30
Cats	2	12	11	25
No Pets	1	3	6	10
Total	11	30	24	65

1. Complete the table assuming all students answered both questions.

2. What fraction of total students have one sibling and cats?

$$\frac{12}{65}$$

3. Does the fraction in #2 represent a marginal or joint frequency? Explain.

Joint frequency because it is found in the center and is a ratio of a part to the total.

4. What fraction of students with 2 or more siblings have dogs?

$$\frac{7}{24}$$

5. Does the fraction in #4 represent a marginal or joint frequency? Explain.

Joint frequency since it is the ratio of a part (7) to a total (24).

Coach Jenkins would like to know if there is any association between the height of the basketball players in his league and the height of their parents. He made a segmented bar graph from the information in the two-way table. Use this information to complete questions 6-10.

	Players under 6 feet	Players 6 feet & over	Total
Both Parents under 6 feet	34	38	72
Both Parents 6 feet & over	18	30	48
Total	52	68	120

6. What fraction of total basketball players surveyed have parents 6 feet and over?

$$\frac{48}{120} = \frac{2}{5}$$

7. What fraction of basketball players under 6 feet have both parents that are 6 feet and over?

$$\frac{18}{52} = \frac{9}{26}$$

8. What fraction of basketball players 6 feet and over have both parents that are 6 feet and over?

$$\frac{30}{68} = \frac{15}{34}$$

9. Explain whether the fractions in questions #6, 7, and 8 marginal or joint frequencies?

They are joint frequencies.

10. What possible associations can Coach Jenkins make about the height of players in his league to the height of their parents? **Answers may vary.**
There is a strong association between players that are under 6 ft and parents height, however there does not seem to be any association between players 6 ft and over.